SUJITH POLISETTY

Physical Design Engineer

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PROFILE SUMMARY

Entry-level VLSI Physical Design Engineer with hands-on training in the ASIC RTL-to-GDSII flow, including floorplanning, placement, clock tree synthesis (CTS), routing, static timing analysis (STA), and timing closure. Familiar with Synopsys ICC2, Fusion Compiler, PrimeTime, and Design Compiler. Experienced in academic and training projects involving RISC-V processor and 1×3 router physical design, timing analysis, and physical verification.

PROFESSIONAL TRAINING

Advanced Physical Design Trainee | Maven Silicon Softech Pvt Ltd., Bangalore **MAY 2025 - Present**

- I Trained in complete ASIC Physical Design flow from netlist to GDSII.
- Gained hands-on experience in floorplanning, CTS, routing, and timing closure using industry-standard EDA tools.
- Developed a solid understanding of trade-offs in timing, power, and area optimization, which are crucial for successful tape-outs in deep sub-micron technology nodes.

EDUCATION

- 2021-2025 BE in Electronics and Communication Engineering from Geethanjali Institute of Science & Technology - **7.7**
- 2019-2021 Intermediate from Krishna Chaitanya Junior College - **8.0**
- 2018-2019 Secondary School from New Little Flowers EM High School - **8.5**

SKILLS

- **Physical Design :-** Floorplan, Placement, Clock Tree synthesis, Routing, Timing Closure, DRC/LVS.
- **EDA Tools :-** Fusion Compiler (Synopsys), PrimeTime (Synopsys), DC & ICCII (Synopsys), Library Manager (Synopsys)
- **VLSI Design Skills :-** Verilog HDL (Basics), Synthesis, Static Timing Analysis, TCL Scripting
- **Soft Skills :-** Time Management, Problem Solving, Video Editing, Adobe Photoshop

PROJECTS

RISC-V
Technology: 32nm
Tool : DC + ICC2, Library manager
Role: Designed and implemented two custom macros into a RISC -V architecture
Description: Implemented the physical design flow for a 32nm RISC-V processor, focusing on timing closure and physical optimization. Analyzed and resolved setup timing violations, optimized placement and routing, reduced signal-wire length by 10%, and performed DRC/LVS verification to check physical design compliance.

ROUTER 1*3
Technology: 32nm
Tool : Fusion Compiler
Role: Designed the whole project in single power domain
Description: Implemented the RTL-to-physical-design flow for a 1×3 router, focusing on floorplanning, placement optimization, and routing. Analyzed timing and physical design reports, investigated design rule violations, and evaluated implementation quality to achieve an optimized physical layout.

INTERNSHIP

RAVEN SoC – Physical Design
Technology: 14nm
Tools: Synopsys Design Compiler (DC), ICC2, Library Manager
Role: Physical Design Engineer
Description: Worked on Raven SoC physical design, focusing on placement optimization and routing. Analyzed timing reports and resolved setup and hold violations. Performed physical verification to ensure design rule compliance.

Floating Point Unit (FPU)
Technology: 14nm
Tool : Synopsys Design Compiler (DC), ICC2, Library Manager
Role: Physical Design Engineer
Description: Implemented the physical design flow for an FPU, focusing on floorplanning, placement, and routing. Analyzed timing reports to identify violations and optimized the layout to meet timing and physical design requirements.